

			Molecular Approaches to Clinical Microbiology in Africa 13-19 September 2025, MRC Unit, The Gambia					  									
PRECOURSE ON LMS - Bioinformatics 1: Introduction to Linux Bioinformatics 1: R Programming for Biologists		Saturday 13 September	Sunday 14 September	Monday 15 September	Tuesday 16 September	Wednesday 17 September	Thursday 18 September	Friday 19 September									
		Transfer: 08:45	Transfer: 07:45	Transfer: 07:45	Transfer: 07:45	Transfer: 07:45	Transfer: 07:45	Transfer: 07:45									
	08:30		Lab safety induction Shola Thomas / Dam Khan	Molecular testing for TB Lizma Streicher, Anzaan Dippenaar, Noutin Michodigni, Shola Thomas, Estelle and Binta	Multi-pathogen whole genome sequencing approaches using Oxford Nanopore Technologies Kanny Diallo, Simon Amoikon, Lizma, Anzaan, Noutin, Dam, Shola, Leonard, Estelle and Binta	Real-time multiplex PCR for amplication and simultaneous detetction of multiple targets Dam Khan, Brenda Kwambana, Kanny Diallo, Simon Amoikon, Shola Thomas, Estelle and Binta	Getting started with R: Setup and first data Leonard Ndwiga, Dam Khan, Shola Thomas	Data wrangling and transformation in R Leonard Ndwiga, Dam Khan, Shola Thomas	08:30								
	08:45								08:45								
	09:00								09:00								
	09:15								09:15								
	09:30	Registration	Germs to genes Martin Maiden	Break						09:30							
	09:45									09:45							
	10:00									10:00							
	10:15									10:15							
	10:30	Welcome and introductions (Wellcome Connecting Science, Training Team, Participants)	Break		Break	Break	Break	Break	Break	10:30							
	10:45									10:45							
	11:00									11:00							
	11:15									11:15							
	11:30	Course Overview Brenda Kwambana, Martin Maiden	Introduction to bioinformatics: Linux and BLAST Keith Jolley Simon Amoikon	Targeted sequencing for detection of drug resistance in TB Lizma Streicher, Anzaan Dippenaar, Noutin Mitchodigni, Shola Thomas, Estelle and Binta	Multi-pathogen whole genome sequencing approaches using Oxford Nanopore Technologies Kanny Diallo, Simon Amoikon, Lizma, Anzaan, Noutin, Dam, Shola, Leonard, Estelle and Binta	Real-time multiplex PCR for amplication and simultaneous detetction of multiple targets Dam Khan, Brenda Kwambana, Kanny Diallo, Simon Amoikon, Shola Thomas	Exploring and cleaning data with R Leonard Ndwiga, Dam Khan, Shola Thomas	Data visualisation in R Leonard Ndwiga, Dam Khan, Shola Thomas		11:30							
	11:45									11:45							
	12:00									Advanced Learning Skills (ALS) Wellcome Connecting Science						Comparing molecular and sequence data for TB Lizma Streicher, Anzaan Dippenaar, Noutin Michodigni	12:00
	12:15																12:15
	12:30	12:30															
	12:45	12:45															
	13:00	Lunch	Lunch	Lunch	Lunch	Lunch	Lunch	Lunch	Action Planning and Wrap Up Wellcome Connecting Science	13:00							
	13:15									13:15							
	13:30									13:30							
	13:45									13:45							
	14:00	Advanced Learning Skills (ALS) Wellcome Connecting Science	Molecular approaches for detecting encapsulated bacteria Dam Khan, Brenda Kwambana, Simon Amoikon, Shola Thomas, Estelle and Binta	Introduction to Phylogenetics Martin Maiden, Keith Jolley, Simon Amoikon	Sequence typing with PubMLST Keith Jolley Simon Amoikon Kanny Diallo	Biobank and sequencing facility tour	Multi-pathogen whole genome sequence analysis Lizma, Anzaan, Noutin, Leonard, Kanny, Simon, Shola, Dam	Lunch		14:00							
	14:15									14:15							
	14:30									14:30							
	14:45									14:45							
	15:00	Sample Management workshop Aaron Dean, MRC biobank team						Transport to Hotel		15:00							
	15:15									15:15							
	15:30									15:30							
	15:45									15:45							
	16:00	Introduction to Genomic Technologies Abdul Karim Sesay	Interpretation of molecular tests for encapsulated bacteria Dam Khan, Brenda Kwambana, Shola Thomas, Simon Amoikon	Break			Multi-pathogen whole genome sequence analysis Lizma, Anzaan, Noutin, Leonard, Kanny, Simon, Shola, Dam		Key	16:00							
	16:15									16:15							
	16:30									16:30							
	16:45									16:45							
	17:00	Technology talks (Illumina, Oxford Nanopore Technologies)		Genes to genomes Martin Maiden	Germs, genes, and genomes Martin Maiden, Keith Jolley	Fireside discussion Brenda Kwambana			Interactive sessions	16:45							
	17:15									17:00							
	17:30									17:15							
	17:45									17:30							
	17:50	Networking drinks and dinner	Seminar - tbc						Laboratory	17:45							
	18:00									18:00							
	18:15									18:15							
18:30	18:30																
19:00		Dinner at the hotel	Dinner at the hotel	Dinner at local restaurant	Dinner at the hotel	Dinner at the MRC			19:00								